CLASS 9
1.3- ETHICS & MORALITY

Fill in the Blanks / Short Answers

1. The guiding principles to decide what is good or bad is known as ____.

Answer: Ethics

2. When building AI solutions, we need to ensure that they follow ____.

Answer: Human Rights

3. Praneet has taken extra packets of mouth freshener after dinner from a restaurant. Is it considered as theft? Is it – Moral or Ethical concern?

Answer: Moral concern – it is a personal belief about right/wrong behaviour, not a formal guiding principle.

4. Rakshit and Aman discuss mobile features; Aman starts getting notifications of mobile models matching his requirement. Which ethical concern does this depict?

Answer: Privacy concern – his conversation/data appears to have been tracked to show targeted notifications/ads.

5. "Preference for one over the other" is known as ____.

Answer: Bias

6. Artificial Intelligence and machine learning systems can display unfair behaviour if not trained properly. (True/False)

Answer: True

7. Search for images of 'personal secretary' on Google, displaying predominantly the images of Women, is an example of ____.

Answer: Bias (gender bias in AI results)

8. An Ethical AI framework makes sure that transparency, fairness and accountability is developed into the systems to provide unbiased results. (True/False)

Answer: True

Answer the Following

1. Differentiate between Ethics and Moral with suitable examples.

Answer: Morals are beliefs dictated by society and can differ across cultures (e.g., "always speak the truth"). Ethics are guiding principles a person chooses to decide right from wrong (e.g., "is it good to speak the truth in all situations?"). Morals are fixed beliefs; Ethics involve questioning them.

2. Define principles of AI.

Answer: AI Ethics Principles are guidelines ensuring AI solutions are safe and fair. The four main principles are: Human Rights (not violating freedom/dignity), Bias (avoiding unfair preference), Privacy (protecting personal data), and Inclusion (not excluding any group).

3. Explain Data Privacy.

Answer: Data privacy means protecting people's personal information collected by AI systems, such as browsing history, photos, or voice data. It ensures AI only collects necessary data, informs users about its use, and keeps that data safe from misuse or breaches.

4. Craft a description of how considerations for inclusivity are addressed during the development of AI models.

Answer: Inclusivity ensures AI doesn't discriminate against any group. Developers check if the AI leaves out any person, benefits rich and poor equally, is easy to use for everyone, and helps all types of users – making the solution fair and accessible to all.

5. Write Major Issues around AI Ethics.

Answer: Major issues include: violation of Human Rights (loss of freedom/jobs), Bias (unfair treatment due to faulty training data), Privacy breaches (misuse of personal data), and lack of Inclusion (excluding certain groups from benefiting from AI).

6. A company had been working on a secret AI recruiting tool. The machine-learning specialists uncovered a big problem: their new recruiting engine did not like women chefs. The system taught itself that male candidates are preferable. It penalised resumes that included the word "women chef". This lead to the failure of the tool.

a. What aspect of AI ethics is illustrated in the given recruiting-tool scenario?

Answer: This illustrates Bias – the AI recruiting tool showed gender discrimination by preferring male candidates and penalising resumes with "women chef."

b. What could be the possible reasons for the ethical concern identified?

Answer: The AI was trained on historical hiring data that had more male candidates, so it learned this pattern as "preferred." Biased or unbalanced training data led the AI to discriminate against women unintentionally.

7. What according to you is a better approach towards this ethical concern? Justify your answer.

Answer: A balanced approach is better: AI should handle repetitive/tedious tasks while humans upgrade their skills for higher-value work. This way, AI eases workload without causing unemployment, as humans adapt and remain in control rather than being replaced.